

Polished Full-Wave Retraining Campaign Results Report

Polished Full-Wave Retraining Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `polished_dataset_full_wave_retraining_2026_06_22` campaign completed the approved full non-paper model-development retraining batch with `108` completed runs and `0` failed runs.

The accepted campaign output directory is `output/training_campaigns/2026-06-29-15-54-18_polished_dataset_full_wave_retraining_2026_06_22`.

The full-wave campaign scalar leaderboard winner is `te_periodic_gru_sequence_fw`, trained on `polished_dataset` with the `polished_point_v1` schema:

- surface: `fw` / `forward_only` ;
- test MAE: `0.001121` deg ;
- test RMSE: `0.001444` deg ;
- validation MAE: `0.001084` deg ;
- trainable parameters: `157,569` .

The program scalar registry still keeps the earlier polished early-wave `te_periodic_gru_sequence_bw` run as the current scalar program winner because its test MAE is lower at `0.001084` deg . This is a normal campaign closeout. It accepts the completed model-development training batch and synchronizes campaign status. It does not promote a new official compensation candidate through the `TE Curve Verification Pipeline` ; that curve-first verification remains a separate operator-approved workflow.

Campaign Scope

The campaign used the prepared `polished_dataset_full_wave_retraining_2026_06_22` package. It covered `36` model families across `global` , `fw` , and `bw` surfaces for `108` total training configurations.

All accepted runs used the polished input columns:

Field	Role
<code>theta</code>	motor position measured in degrees
<code>theta_dot</code>	motor velocity derived from position
<code>tau_load</code>	applied load in Nm
<code>T</code>	oil temperature
<code>theta_TE</code>	measured transmission error target

Included model families

Family A	Family B	Family C
<code>feedforward</code>	<code>gru_sequence</code>	<code>harmonic_regression</code>
<code>lstm_sequence</code>	<code>periodic_gru_sequence</code>	<code>periodic_lstm_sequence</code>

Family A	Family B	Family C
periodic_mlp	periodic_mlp_harmonic	periodic_temporal_convolution
residual_harmonic_gru_sequence_dense240	residual_harmonic_gru_sequence_dense360	residual_harmonic_gru_sequence_sparse_rcim
residual_harmonic_lstm_sequence_dense240	residual_harmonic_lstm_sequence_dense360	residual_harmonic_lstm_sequence_sparse_rcim
residual_harmonic_mlp	temporal_convolution	tree
wave3_1_sequential_residual_offset_probe	wave3_2_clean_sequential_residual_offset	wave3_2_harmonic_residual_offset
wave3_3_curve_aware_pointwise_control	wave3_3_full_curve_composite	wave3_3_raw_centered_shape_curve_aware
wave3_3_raw_offset_curve_aware	wave4_1_log_cosh_robust_losses	wave4_1_mae_robust_loss
wave4_1_smooth_l1_robust_loss	wave4_2_gaussian_nll	wave4_2_quantile_p10_p50_p90
wave4_3_mixture_density_k2	wave4_3_mixture_density_k3	wave4_4_causal_tcn_latent_offset_residual
wave4_4_gru_latent_offset_residual	wave5_1_harmonic_prior_pointwise_control	wave5_1_harmonic_prior_smooth_l1_structured

Queue And Artifact Audit

Check	Result
Completed queue entries	108
Failed queue entries	0
Running queue entries	0
Pending queue entries	0
Leaderboard entries	108
Dataset entries recorded as polished_dataset	108
Schema entries recorded as polished_point_v1	108
Missing referenced metrics/reports/checkpoints	0

The campaign completed cleanly. All accepted leaderboard rows use the four polished input features `theta` , `theta_dot` , `tau_load` , and `T` , with `theta_TE` as the single target.

Surface Winners

Surface	Best Run	Family	Test MAE	Test RMSE	Val MAE	Params
Global	te_periodic_gru_sequence_global	periodic_gru	0.001159	0.001465	0.001132	157,569
Forward-only	te_periodic_gru_sequence_fw	periodic_gru	0.001121	0.001444	0.001084	157,569
Backward-only	te_periodic_gru_sequence_bw	periodic_gru	0.001166	0.001481	0.001158	157,569

Scalar Leaderboard Snapshot

The table below shows the first 12 entries from the scalar leaderboard. The full ordered list, with exact run names and artifact paths, is stored in `output/training_campaigns/2026-06-29-15-54-18_polished_dataset_full_wave_retraining_2026_06_22/campaign_leaderboard.yaml` .

Rank	Compact Run	Surface	Family	Test MAE	Test RMSE	Val MAE
1	periodic_gru_fw	fw	periodic_gru	0.001121	0.001444	0.001084
2	periodic_gru_global	global	periodic_gru	0.001159	0.001465	0.001132
3	periodic_gru_bw	bw	periodic_gru	0.001166	0.001481	0.001158
4	periodic_lstm_bw	bw	periodic_lstm	0.001226	0.001558	0.001230
5	periodic_mlp_harmonic_global	global	periodic_mlp_harmonic	0.001309	0.001794	0.001265
6	periodic_mlp_harmonic_bw	bw	periodic_mlp_harmonic	0.001342	0.001807	0.001188
7	periodic_mlp_harmonic_fw	fw	periodic_mlp_harmonic	0.001360	0.001845	0.001209
8	wave4_3_mdn_k3_global	global	wave4_3_mixture_density_k3	0.001544	0.001992	0.001407
9	periodic_lstm_fw	fw	periodic_lstm	0.001555	0.001983	0.001513
10	periodic_lstm_global	global	periodic_lstm	0.001601	0.002029	0.001536
11	wave4_3_mdn_k3_fw	fw	wave4_3_mixture_density_k3	0.001671	0.002181	0.001501
12	feedforward_bw	bw	feedforward	0.001686	0.002175	0.001630

Registry And Program Effects

The completed run updated family-level registries for the full polished model bank and refreshed the program scalar registry timestamp.

The campaign-level scalar winner is `te_periodic_gru_sequence_fw` , but the current program-level scalar winner remains the earlier polished early-wave `te_periodic_gru_sequence_bw` run because it has the lower test MAE (`0.001084 deg` versus `0.001121 deg`).

This scalar result is useful as a retraining result, but it does not replace the official direction-parallel curve-verified leaders until a separate `TE Curve Verification Pipeline` refresh is run and accepted. The repository must keep `global` , `Fw` , and `Bw` surfaces visible as separate decision surfaces.

Acceptance Decision

The full-wave campaign is accepted as a completed polished-dataset retraining batch because:

- all `108` selected configs completed;
- no queue entries remain pending, running, or failed;
- every leaderboard entry records `polished_dataset` ;
- every leaderboard entry records `polished_point_v1` ;
- every accepted model uses four input features and one TE target;
- every referenced metric, training report, and checkpoint artifact exists.

Closeout Notes

This closeout supersedes the earlier early-wave parallel training closeout as the latest normal model-development campaign closeout. The early-wave and RCIM polished closeouts remain preserved as previous completed campaign evidence.

The `TE Program Status And Closeout Ledger` requires a content update because the latest normal campaign closeout changed from the early-wave parallel batch to the full-wave retraining campaign. The official curve-verified leaders do not change in this closeout because the `TE Curve Verification Pipeline` was not run.

Follow-Up

Recommended next steps

1. prepare a separate operator-run `TE Curve Verification Pipeline` refresh that includes the polished RCIM, early-wave, and full-wave candidate set;
2. keep the scalar full-wave winner visible as campaign evidence, but use the curve-first policy for promotion;
3. evaluate whether the full-wave retraining results should feed a compact TwinCAT-facing candidate export set after curve verification.